

EE1060: Differential Equations and Transform Techniques - Quiz 2

Name: _____

Roll No: _____

Instructions

1. The quiz will last for 45 minutes, beginning at 14:45. Please ensure that you submit your paper by 15:30. **Late submissions may incur penalties**, so make sure to return the paper on time.
2. Go through all the questions carefully. Any **misinterpretations** will **not** be considered.
3. Write all your answers in the answer book. It is essential that your responses are **neat** and **legible**. If your answers are **not clear or readable**, they will **not be graded**. You cannot later claim that there are answers written in some obscure corner of the paper that were not evaluated. Ensure your answers are clearly presented and easily identifiable.
4. After answering all the questions you wish to, **mark the end of your answer book clearly with a distinct mark, such as a line.**

-
1. (2 points) Solve the first-order ordinary differential equation

$$(x + xy) dx + (x^2) dy = 0 \quad (1)$$

2. (2 points) Solve the first-order ordinary differential equation

$$\frac{dy}{dx} - 5y = e^{-2x}y^2 \quad (2)$$

3. (3 points) Consider the ODE

$$\frac{dy}{dt} = y^3 - 6y^2 + 9y \quad (3)$$

- (a) (1 point) Compute the equilibrium points of the system.
 - (b) (1 point) Derive the ODE corresponding to linearized system around the any one equilibrium point computed in 3(a) .
 - (c) (1 point) Derive the condition on step length for with Runge-Kutta 2nd order method is numerically stable.
4. (3 points) Consider the ODE

$$\frac{dy}{dt} = y(2 - y)(y - \mu) \quad (4)$$

- (a) (1 point) For $\mu = 1$, draw the phase plane trajectories.
- (b) (2 points) Plot the bifurcation diagram for the system.

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-
1. (2 points) Solve the first-order ordinary differential equation

$$(x + xy) dx + (x^3) dy = 0 \quad (1)$$

2. (2 points) Solve the first-order ordinary differential equation

$$\frac{dy}{dx} - 6y = e^{-2x}y^2 \quad (2)$$

3. (3 points) Consider the ODE

$$\frac{dy}{dt} = y^3 - 6y^2 + 8y \quad (3)$$

- (a) (1 point) Compute the equilibrium points of the system.
 - (b) (1 point) Derive the ODE corresponding to linearized system around the any one equilibrium point computed in 3(a) .
 - (c) (1 point) Derive the condition on step length for with Runge-Kutta 2nd order method is numerically stable.
4. (3 points) Consider the ODE

$$\frac{dy}{dt} = y(3 - y)(y - \mu) \quad (4)$$

- (a) (1 point) For $\mu = 1$, draw the phase plane trajectories.
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-
1. (2 points) Solve the first-order ordinary differential equation

$$(x + xy) dx + (x^4) dy = 0 \quad (1)$$

2. (2 points) Solve the first-order ordinary differential equation

$$\frac{dy}{dx} - 4y = e^{-2x}y^2 \quad (2)$$

3. (3 points) Consider the ODE

$$\frac{dy}{dt} = y^3 - 3y^2 + 2y \quad (3)$$

- (a) (1 point) Compute the equilibrium points of the system.
 - (b) (1 point) Derive the ODE corresponding to linearized system around the any one equilibrium point computed in 3(a) .
 - (c) (1 point) Derive the condition on step length for with Runge-Kutta 2nd order method is numerically stable.
4. (3 points) Consider the ODE

$$\frac{dy}{dt} = y(3 - y)(y - 2\mu) \quad (4)$$

- (a) (1 point) For $\mu = 1$, draw the phase plane trajectories.
- (b) (2 points) Plot the bifurcation diagram for the system.

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-
1. (2 points) Solve the first-order ordinary differential equation

$$(x + xy) dx + xdy = 0 \quad (1)$$

2. (2 points) Solve the first-order ordinary differential equation

$$\frac{dy}{dx} - 3y = e^{-2x}y^2 \quad (2)$$

3. (3 points) Consider the ODE

$$\frac{dy}{dt} = y^3 - 4y^2 + 4y \quad (3)$$

- (a) (1 point) Compute the equilibrium points of the system.
 - (b) (1 point) Derive the ODE corresponding to linearized system around the any one equilibrium point computed in 3(a) .
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$$\frac{dy}{dt} = y(2 - y)(y - 2\mu) \quad (4)$$

- (a) (1 point) For $\mu = 1$, draw the phase plane trajectories.
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